

ATIVIDADE DADO ELETRÔNICO

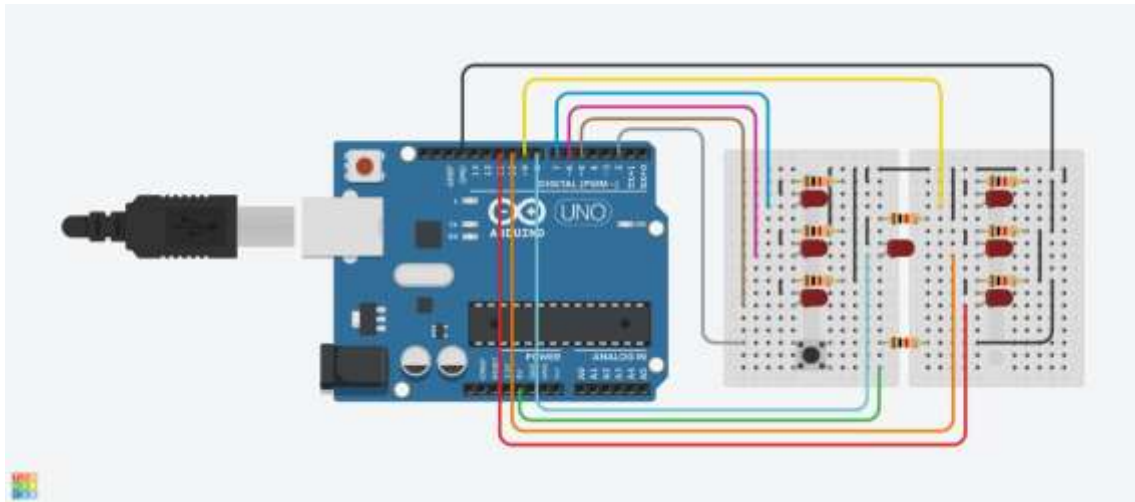
Sistemas Embarcados e IoT

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Dado Eletrônico –



Código –

```
// C++ code
//
int pinoLed1 = 7; int
pinoLed2 = 6; int
pinoLed3 = 5; int
pinoLed4 = 8; int
pinoLed5 = 9; int
pinoLed6 = 10; int
pinoLed7 = 11; int
pinoBotao = 2;

void setup() {
  pinMode(pinoLed1,OUTPUT);
```

```
pinMode(pinoLed2,OUTPUT);
pinMode(pinoLed3,OUTPUT);
pinMode(pinoLed4,OUTPUT);
pinMode(pinoLed5,OUTPUT);
pinMode(pinoLed6,OUTPUT);
pinMode(pinoLed7,OUTPUT);
pinMode(pinoBotao,INPUT_PULLUP);
randomSeed(analogRead(0));
}
```

```
void loop() {
if(digitalRead(pinoBotao) == HIGH) {
    sorteia();
}
}
```

```
void sorteia() {
    int numeroSorteado = random(1,8);

    digitalWrite(pinoLed1, LOW);
    digitalWrite(pinoLed2, LOW); digitalWrite(pinoLed3,
LOW); digitalWrite(pinoLed4, LOW);
    digitalWrite(pinoLed5, LOW); digitalWrite(pinoLed6,
LOW); digitalWrite(pinoLed7, LOW);

    if(numeroSorteado==1)
    {
        digitalWrite(pinoLed4,HIGH);
    }
}
```

```
}
```

```
if(numeroSorteado==2)
```

```
{
```

```
    digitalWrite(pinoLed1, HIGH);
```

```
    digitalWrite(pinoLed7, HIGH);
```

```
}
```

```
if(numeroSorteado==3)
```

```
{
```

```
    digitalWrite(pinoLed1, HIGH);
```

```
    digitalWrite(pinoLed4, HIGH);
```

```
    digitalWrite(pinoLed7, HIGH);
```

```
}
```

```
if(numeroSorteado==4)
```

```
{
```

```
    digitalWrite(pinoLed1, HIGH);
```

```
    digitalWrite(pinoLed3, HIGH);
```

```
    digitalWrite(pinoLed5, HIGH);
```

```
    digitalWrite(pinoLed7, HIGH);
```

```
}
```

```
if(numeroSorteado==5)
```

```
{
```

```
    digitalWrite(pinoLed1, HIGH);
```

```
    digitalWrite(pinoLed3, HIGH);
```

```
    digitalWrite(pinoLed5, HIGH);
```

```
        digitalWrite(pinoLed7, HIGH);  
        digitalWrite(pinoLed4, HIGH);  
    }
```

```
if(numeroSorteado==6)  
{  
    digitalWrite(pinoLed1, HIGH);  
    digitalWrite(pinoLed2, HIGH);  
    digitalWrite(pinoLed3, HIGH);  
    digitalWrite(pinoLed5, HIGH);  
    digitalWrite(pinoLed6, HIGH);  
    digitalWrite(pinoLed7, HIGH);  
}
```

```
    delay(3000);  
    digitalWrite(pinoLed1, LOW);  
    digitalWrite(pinoLed2, LOW);  
    digitalWrite(pinoLed3, LOW);  
    digitalWrite(pinoLed4, LOW);  
    digitalWrite(pinoLed5, LOW);  
    digitalWrite(pinoLed6, LOW);  
    digitalWrite(pinoLed7, LOW);  
}
```